Christopher Hynes

646-844-5242 | resume@chynes.org | LinkedIn | Personal Website

Education

Georgia Institute of Technology Jan. 2024 – Expected Dec. 2026
M.S. in Computer Science; Artificial Intelligence specialization; GPA: 4.0 *Atlanta, GA*

University of South Carolina: Honors College Aug. 2018 – Aug. 2022
B.S. in Statistics and Economics; Summa cum laude, GPA: 4.0 *Columbia, SC*

Universidad Carlos III de Madrid Sep. 2021 – Jun. 2022
Study Abroad Program *Madrid, Spain*

Technical Skills

Programming: Python, R, SQL; familiar with Java, Stata, SAS
AI Assistants & Agent Tools: Codex, Claude Code, Gemini CLI, Ollama, OpenClaw
Machine Learning & Data: PyTorch, scikit-learn, TensorFlow, Hugging Face Transformers, pandas, NumPy, tidyverse; supervised and unsupervised learning, reinforcement learning, NLP, causal inference
Development & Tools: Git, Linux, Bash, Docker, Jupyter, Colab, Streamlit, Astro, Quarto, GitHub Actions, LaTeX

Research & Professional Experience

Data Analyst Intern Jun. 2023 – Sep. 2023
GreyNoise Intelligence; Finance and Operations Team *Washington, DC*

- Analyzed customer churn in R; subscribers who had not used a free trial were twice as likely to churn, informing a greater emphasis on free trials
- Built a Streamlit dashboard to communicate churn and lead-conversion findings; developed ARR-weighted reporting metrics
- Automated Excel data-quality checks and built vendor/transaction reports; partially automated industry classification using HubSpot NAICS data
- Developed Excel models for mid-term subscription upgrades and co-term billing calculations for the customer experience team
- Built a cash investment tracker that parsed financial-institution data and incorporated previously omitted interest income into monthly forecasts

Undergraduate Research Assistant May 2021 – Oct. 2021
Darla Moore School of Business; Division of Research *Columbia, SC*

- Reviewed literature on spatial segregation metrics and neighborhood effects in Charleston, SC
- Cleaned and analyzed MSA-level economic, social, and demographic data to study STEM worker agglomeration
- Calculated location quotients and ran regressions in Stata/R to identify drivers of STEM clustering across MSAs

Undergraduate Research Assistant Jan. 2020 – Jan. 2021
University of South Carolina; Department of Economics *Columbia, SC*

- Digitized and organized CDC vital statistics data to support analysis of racial disparities in birth outcomes post-hospital desegregation

Selected Technical & Research Projects

Signal Assistant | *Python, Ollama, signal-cli, Docker* Jan. – Mar. 2026

- Built a self-hosted Signal assistant with local Ollama inference, configurable personas, per-chat memory, rolling summaries, and administrative controls
- Bounded concurrent inference and coordinated shared transport access while preserving message order; added automated tests with fake clients

Multi-LLM: Best-of-N Response Selection | *Python, FastAPI, Ollama, Docker* May 2026

- Built an OpenAI-compatible best-of-N proxy with concurrent candidate generation, separate LLM judging, partial-failure handling, and deterministic fallback

AI Safety & Governance News Tracker | *Python, Streamlit* Jul. 2025

- Built a daily news pipeline with deduplication, VADER sentiment, and phrase-frequency analysis; created a Streamlit dashboard with interactive trends and article filters

Research Portfolio Website | *Astro, TypeScript* Oct. 2026

- Built a research portfolio with interactive model-comparison figures, searchable case studies, and accessible data tables
- Automated deployment, link validation, and browser and accessibility testing through GitHub Actions

Graduate Machine Learning Studies | *Python, PyTorch, scikit-learn* Sep. 2025 – Nov. 2025

- Developed and debugged ML pipelines across four graduate projects in supervised learning, unsupervised learning, optimization, and reinforcement learning
- Compared model performance, stability, and computational cost through hyperparameter searches, repeated trials, and statistical tests on datasets containing up to 7.7 million records
- Authored four technical reports interpreting algorithm behavior, computational tradeoffs, and experimental limitations

Hospital Readmission Replication and Extension | *Python, PyTorch* Mar. 2025 – Apr. 2025

- Implemented preprocessing, the GRU baseline, Word2Vec embeddings, and evaluation in a collaborative PyTorch replication of CONTENT on 3,000 synthetic patients
- Conducted validation-based tuning experiments using a teammate's grid-search module, ran 30-trial comparisons, and co-authored the report
- Found no significant CONTENT advantage in the initial configuration; after tuning, reported PR-AUC 0.6464 vs. 0.6391 for GRU ($p < 0.0001$)

ML Trading Strategy and Market Simulator | *Python, pandas, NumPy* Jan. 2024 – May 2024

- Built a bagged random-tree learner and market simulator; compared learned, manual, and buy-and-hold strategies on out-of-sample data

Honors Thesis: The Effects of Chile's Neoliberal Reforms | *R, tidyverse* Aug. 2020 – May 2022

- Constructed a synthetic-control counterfactual from World Bank data to estimate the effect of Chile's reforms on GDP per capita
- Estimated 2019 GDP per capita to be 76% above the counterfactual; tested robustness across alternative donor pools, predictors, and a placebo study

HCI Study: Streamlined Camera Mode Selection in iOS | *Figma* Oct. 2024 – Nov. 2024

- Used observation, a 28-response survey, concept comparison, and Figma prototyping; six final-stage participants rated perceived efficiency and intuitiveness higher than standard iOS menus

Selected Coursework

Computer Science: Machine Learning, Natural Language Processing, Big Data Analytics, Computer Networks, Knowledge-Based AI, Algorithmic Design, Software Development Process, Human-Computer Interaction
Statistics & Economics: Applied Multivariate Statistics, Statistical Inference, Mathematical Statistics, Probability, Introductory Econometrics, Quantitative Microeconomics

Awards & Honors

- **Stephen D. Durham Award**: Awarded to the top senior statistics student (2022)
- **Phi Beta Kappa** (2022) | **Palmetto Fellows Scholarship** (2018–2022) | **Presidential Scholars Award** (2018–2022)

Languages

English: Native | Spanish: Advanced | German: Intermediate | French: Basic